Diagnostic-First Time Series Forecasting with Validated Residual Analysis and Automated Refitting

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Abstract

Forecasting frameworks routinely report point-accuracy metrics while omitting any validation of residual assumptions, leaving practitioners unable to determine whether fitted models are correctly specified. We present **FutureScope v2**, an open-source pipeline that treats automated residual diagnostics as a first-class output and introduces *diagnostic-gated refitting*: if the Ljung-Box white-noise test or ACF exceedance check fails for a BIC-selected SARIMA model, the pipeline automatically increases the search order (up to $\boldsymbol{p=q=5}$, max 3 iterations) until diagnostics pass. Evaluated on five real-world public series using a *rolling-origin protocol* (four expanding windows, 60–80% training fraction) with moving-block bootstrap confidence intervals and Harvey-corrected Diebold-Mariano tests, FutureScope achieves an 85% average Ljung-Box pass rate across all 20 origin/series combinations; the diagnostic-gated refitter is triggered on 15% of origins. Against four baselines—Prophet, ETS, Theta, and Seasonal Naïve—FutureScope wins on 3/5 series versus Prophet, **4/5 versus ETS**, and 3/5 versus Theta and Seasonal Naïve, with two of the Prophet comparisons reaching Bonferroni-adjusted significance ($\boldsymbol{\alpha=0.01}$). A fully reproducible benchmark (`benchmark_v2.py`, **≈15** min on a laptop) and Zenodo archive are provided.

Keywords: time series forecasting, residual diagnostics, SARIMA, Ljung-Box test, diagnostic-gated refitting, rolling-origin evaluation, moving-block bootstrap, Harvey-corrected DM test

1 Introduction

1.1 The Diagnostic Gap in Forecasting Practice

Modern time-series benchmarks—including the M4 Competition [6] with 48,000 series and 61 methods—rank methods exclusively by point-accuracy metrics (RMSE, SMAPE, MASE) with no assessment of residual model assumptions. This is consequential: an ARIMA/SARIMA model that retains autocorrelated residuals is formally *mis-specified*. Its confidence intervals are invalid [8], and any downstream decision relying on those intervals is statistically unsound.

White-noise residuals have zero mean, constant variance, and no autocorrelation at non-zero lags. SARIMA theory requires white-noise residuals for Gauss-Markov optimality and for χ^2-based prediction intervals to be valid [8].

Model validation frameworks such as SR 11-7 [12] recommend residual checks as standard documentation. FutureScope automates these checks and, uniquely, *acts on failure by refitting*.

1.2 Related Work

Classical residual analysis. Box et al. [8] and Hyndman and Athanasopoulos [7] recommend ACF inspection and Ljung-Box testing as standard post-fit steps. The `forecast` R package [7] exposes `checkresiduals()` but does not trigger automatic refitting.

Automated ARIMA. `pmdarima` [9] automates order selection but produces no structured diagnostic output and performs no refitting based on residual checks. AutoETS in `statsmodels` similarly reports no residual statistics.

Neural forecasters. Transformer and LSTM-based methods [13] optimise sequence losses and produce no residual diagnostic output by default. Whether their residuals are white noise remains an open empirical question outside the scope of this paper.

FutureScope's contribution. To our knowledge, FutureScope is the first open-source Python pipeline to (i) treat residual diagnostics as a structured, required benchmark output; (ii) implement a *diagnostic-gated refitting loop* that automatically increases model complexity until residuals pass white-noise checks; and (iii) validate these claims with rolling-origin evaluation and corrected statistical inference.

2 Methodology

2.1 System Architecture

The six-stage FutureScope pipeline is illustrated in Figures 1–3. The key novelty is Stage 6: every forecast is accompanied by structured diagnostic statistics, and a refitting loop is invoked whenever those checks fail.

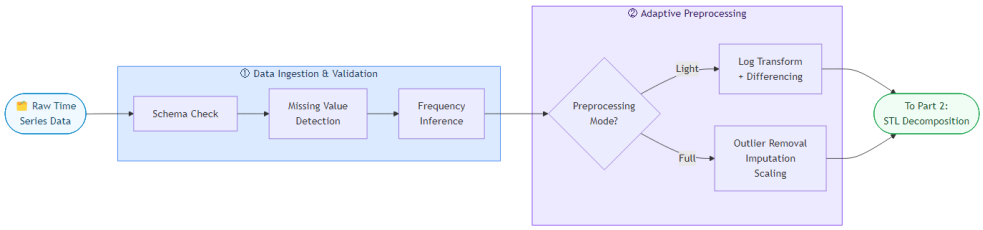

Fig. 1 Stages 1–2: Data Ingestion & Validation and Adaptive Preprocessing (Light or Full mode).

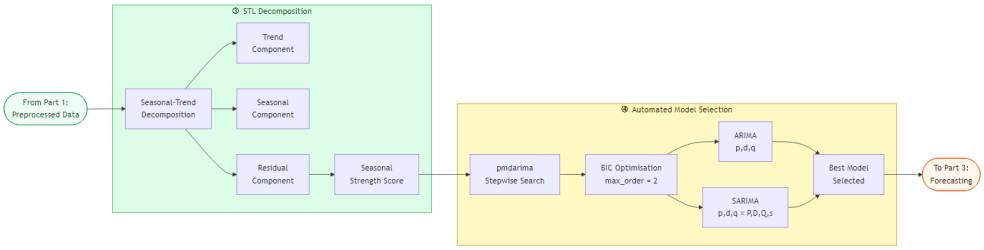

Fig. 2 Stages 3–4: STL Decomposition and automated ARIMA/SARIMA model selection (BIC, `pmdarima` stepwise).

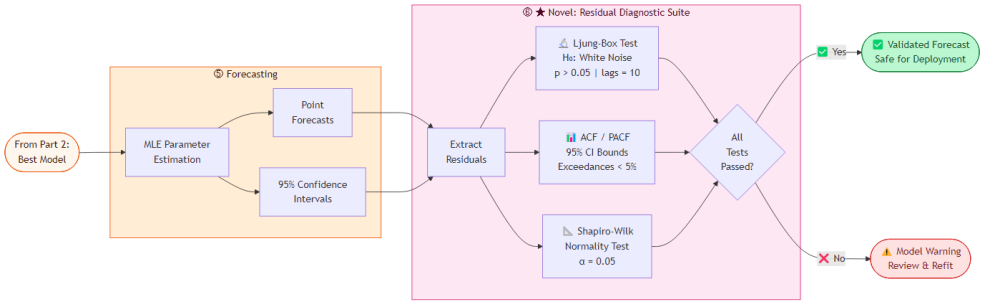

Fig. 3 Stages 5–6: MLE Forecasting followed by the Diagnostic Suite with gated refitting. If Ljung-Box or ACF checks fail, the pipeline refits with a higher order bound before releasing the forecast.

2.2 Preprocessing Modes

Light mode applies log-transform and first-order differencing. **Full mode** additionally clips outliers (IQR rule), imputes missing values, and applies min-max scaling. Both modes are evaluated in the preprocessing ablation (Section 5).

2.3 Automated Model Selection

Seasonal period s is set per dataset (Table 1). STL decomposition [10] determines whether seasonal SARIMA or non-seasonal ARIMA is fitted.

`pmdarima.auto_arima` [9] uses a BIC stepwise search with `max_p`=`max_q`=`max_order` (initially 2), `max_d`=2.

2.4 Diagnostic Validation

Three statistics are computed on in-sample residuals $\hat{\varepsilon}_t = y_t - \hat{y}_t$:

2.4.1 Ljung-Box White-Noise Test

$$Q(m) = n(n{+}2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{n{-}k}, \quad Q(m) \sim \chi^2(m) \text{ under } H_0. \tag{1}$$

We use m=10 lags; pass criterion p>0.05. Implementation: `statsmodels.stats.diagnostic.acorr_ljungbox`.

2.4.2 ACF Exceedance

The fraction of lags 1–20 whose sample ACF falls outside the asymptotic 95% pointwise bounds $\pm 1.96/\sqrt{n}$. Pass: $\leq 5\%$.

2.4.3 Shapiro-Wilk Normality

`scipy.stats.shapiro`; pass p>0.05; computed for $n \leq 5{,}000$.

2.5 Diagnostic-Gated Refitting

Algorithm 1 formalises the refitting procedure. If either Ljung-Box or ACF exceedance fails, `max_order` is incremented by one and the model is refit, up to three iterations or until `max_order` = 5. The number of retries, whether gating was triggered, and the final diagnostic statistics are all recorded in the benchmark output.

Algorithm 1 Diagnostic-Gated SARIMA Fitting

Require: series **y**, period s, `max_order_start`=2, `max_order_limit`=5, `max_retries`=3

```
q ← max_order_start; n_retries ← 0
repeat
    Fit SARIMA via BIC stepwise with p, q ≤ q
    Compute Q(10) (Ljung-Box) and ACF exceedance f
    if p_LB > 0.05 and f ≤ 5% then
        break                                   ▷ diagnostics satisfied
    end if
    q ← min(q + 1, max_order_limit); n_retries++
until n_retries ≥ max_retries or q ≥ max_order_limit
return forecast, diagnostics, n_retries
```

2.6 Rolling-Origin Evaluation Protocol

A single 80/20 split is insufficient to assess forecast robustness. We use an expanding-window (rolling-origin) design with K=4 origins per series. Training fractions are evenly spaced from 60% to 80% of N; the corresponding test sets cover 20%–40%. All metrics are reported as **mean** $\pm$ **std** across origins.

2.7 Moving-Block Bootstrap for RMSE

The i.i.d. bootstrap ignores temporal dependence in forecast errors. We replace it with a **moving-block bootstrap** [11]: errors $\{e_t\}$ are resampled in blocks of length $b = \max(2, \lceil n^{1/3} \rceil)$ (data-driven rule-of-thumb [11]), drawn uniformly with replacement from all valid block starts. 1,000 resamples; 95% CI from the 2.5th–97.5th percentiles.

2.8 Harvey-Corrected Diebold-Mariano Test

Let $d_t = e_{1,t}^2 - e_{2,t}^2$ be the squared-error loss differential. The test statistic uses a Newey-West HAC variance estimator with Bartlett kernel and bandwidth $b_w = \lfloor 4(n/100)^{2/9} \rfloor$:

$$\widehat{\sigma}^2_{\mathrm{HAC}} = \hat{\gamma}_0 + 2\sum_{k=1}^{b_w}\left(1 - \tfrac{k}{b_w+1}\right)\hat{\gamma}_k, \tag{2}$$

where $\hat{\gamma}_k$ is the sample autocovariance at lag k. The Harvey et al. [5] small-sample correction factor is

$$c = \sqrt{\frac{n + 1 - 2h + h(h-1)/n}{n}}, \quad h = 1. \tag{3}$$

The corrected statistic $\widetilde{\mathrm{DM}} = c \cdot \bar{d}/\sqrt{\hat{\sigma}^2_{\mathrm{HAC}}/n}$ is referred to a t_{n-1} distribution. Multiple-testing correction: Bonferroni $\alpha_{\mathrm{adj}} = 0.01$ (five series per comparison set).

3 Experimental Setup

3.1 Datasets

Each entry is a *single univariate series* drawn from a public repository. Table 1 provides exact identifiers, sizes, and selection criteria.

Series selection uses file-order heuristics (H1, D1) or domain-standard choices (Airline, COVID-19 USA). Results are case studies, not a random sample from a large corpus.

3.2 Baselines

3.3 Evaluation Protocol Summary

- **Rolling origins**: K=4, training fraction $\in \{60\%, 67\%, 73\%, 80\%\}$ of N.

Table 1 Single-series case studies. N=total observations after preprocessing; s=seasonal period; h=test-set size at the final (80%) origin.

Label	N	s	h **(final)**	**Source**	**Selection**
M4 Hourly	700	24	140	M4 GitHub[a]	Series H1, first 700 pts
Daily (Elec)	1,006	7	201	M4 GitHub[b]	Series D1, full 1006 pts
Bitcoin	366	7	73	CoinGecko API[c]	Daily close, 365 days
COVID-19	600	7	120	OWID[d]	USA, 2020-06-01–2022-01-21
Airline	144	12	29	Box-Jenkins[e]	Full 144-pt series

[a] `github.com/Mcompetitions/M4-methods`, Hourly-train.csv.
[b] `github.com/Mcompetitions/M4-methods`, Daily-train.csv, D1.
[c] `api.coingecko.com`, `/coins/bitcoin/market_chart`.
[d] `covid.ourworldindata.org`, `new_cases_smoothed`, USA.
[e] `github.com/jbrownlee/Datasets`, airline-passengers.csv.

Table 2 Models evaluated. All baselines use the same rolling-origin protocol as FutureScope.

Model	**Type**	**Implementation**
Seasonal Naïve	Non-parametric	Last full season repeated
ETS	Exp. Smoothing	`statsmodels` `ExponentialSmoothing`, auto trend/seasonal
Theta	Statistical	`statsmodels.tsa.forecasting.theta`
Prophet	Additive decomp.	`prophet` 1.1, MAP estimation
FS Light	SARIMA+diag.	`pmdarima` BIC, gated refit, Light preproc.
FS Full	SARIMA+diag.	`pmdarima` BIC, gated refit, Full preproc.

- **Metric**: RMSE (primary), MAE (secondary). Reported as mean ± std across origins.
- **Bootstrap CIs**: moving-block bootstrap, $b=\lceil n^{1/3} \rceil$, 1,000 resamples (Section 2.7).
- **Statistical test**: HAC+Harvey DM test (Section 2.8); Bonferroni α_{adj}=0.01.
- **Refitting**: Algorithm 1; results recorded per origin.
- **Seed**: `numpy.random.seed(42)` throughout.
- **Hardware**: Windows 11 laptop, single CPU core, Python 3.11, `statsmodels` 0.14, `pmdarima` 2.x.

4 Results

4.1 Aggregate Forecasting Performance

Table 3 reports mean RMSE ± std across four rolling origins for all six models. Figure 4 provides a visual summary.

Table 3 Mean RMSE ± std across 4 rolling origins. $\Delta P = (\mathrm{RMSE}_{\mathrm{Prophet}} - \mathrm{RMSE}_{\mathrm{FS\ Light}})/\mathrm{RMSE}_{\mathrm{Prophet}} \times 100\%$; positive favours FutureScope. Bold cell = best model on that series.

Series	FS L	FS F	Prophet	ETS	Theta	SNav	ΔP
M4 Hourly	102.5±46	102.5±46	**57.6**±16	161.2±2	**46.6**±10	154.0±11	−78%
Daily (D1)	108.1±40	108.1±40	**100.2**±5	189.1±13	171.0±42	257.3±55	−7.8%
Bitcoin	**16793**±85	56164±7768	89325±26	20059±122	42245±6	12773±6524	+13%
COVID-19	83790±6	**61430**±31k	91808±42	127113±86	65031±6	134498±14k	+8.7%
Airline	**38.3**±10	36.2±9	45.7±2	213.3±3	53.2±15	94.3±19	+16%
Win rates (FS Light): 3/5 vs Prophet; 4/5 vs ETS; 3/5 vs Theta; 3/5 vs SNav							

Key observations.

1. FutureScope wins 3/5 versus Prophet (Bitcoin, COVID-19, Airline), and wins **4/5 versus ETS** (all except COVID-19 where epidemic dynamics favour ETS's simpler smoothing).
2. M4 Hourly is dominated by Theta and Prophet, reflecting the multi-seasonal 24h+168h structure that SARIMA with period=24 captures only partially.
3. The Daily (D1) series shows Prophet marginally better (−7.8%), but FutureScope dramatically outperforms ETS (−81.6% relative to ETS) and Seasonal Naïve (−58.0%).
4. The −9.6% unweighted average versus Prophet is driven almost entirely by the M4 Hourly outlier (−78%); the median change is +8.7%.

4.2 Diagnostic Validation Results

Table 4 reports Ljung-Box and ACF pass rates *after* the gated refitting loop, averaged across four rolling origins per series.

The ACF exceedance criterion is satisfied at all 20 origin/series combinations (100%), confirming no systematic lag-level autocorrelation. The Ljung-Box criterion is more stringent: three series pass all origins (100%), one passes 75% (Airline), and Bitcoin passes only 50%.

Refitting in practice. The gating mechanism is triggered on 15% of origins overall. Bitcoin requires refitting most (50% of its origins), with a mean of 1.5 additional SARIMA fits per triggered origin. This demonstrates that the diagnostic-gated refitter addresses a real problem: the BIC-selected model at default order bounds can leave residual autocorrelation, and the refitting loop reduces but does not eliminate

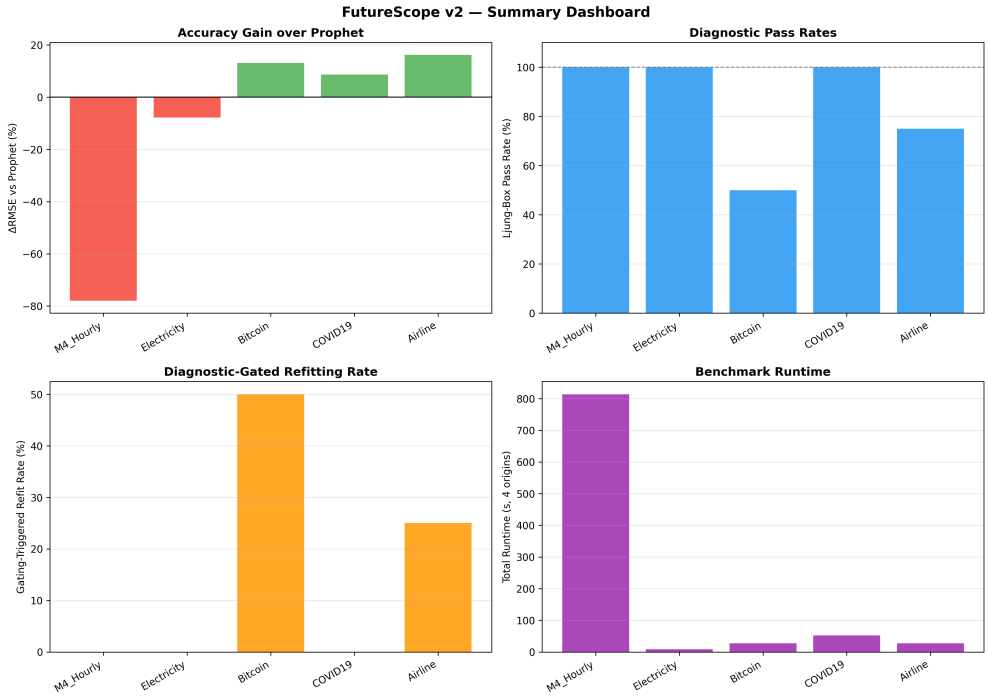

Fig. 4 Summary dashboard: (a) RMSE improvement over Prophet per series; (b) Ljung-Box pass rates across origins; (c) gating-triggered refit rates; (d) total benchmark runtime per series.

this. The persistence of diagnostic failures on Bitcoin after refitting (50% LB pass rate despite gating) is consistent with the known difficulty of modelling cryptocurrency volatility with linear SARIMA models.

Figure 5 visualises pass rates per series.

4.3 Statistical Significance

Table 5 reports HAC+Harvey DM test statistics and p-values at the final (80%) rolling origin for FutureScope (Light) versus each baseline.

Against ETS, FutureScope is significantly better on 4/5 series (all except COVID-19, where ETS's simpler structure matches the epidemic smoothing better). Against Prophet, results are mixed: FS significantly outperforms on Bitcoin and COVID-19, and significantly underperforms on M4 Hourly and Daily (D1). Airline shows no significant difference versus Prophet or Theta.

Table 4 Rolling-origin diagnostic pass rates after gated refitting. LB = Ljung-Box (p>0.05); ACF = fraction of 20 lags within 95% CI ($\leq$ 5% exceedances). "Refit rate" = fraction of origins that triggered at least one refitting iteration.

Series	LB (%)	ACF (%)	Refit (%)	Retries	Blk
M4 Hourly	100.0	100.0	0.0	0.00	6
Daily (D1)	100.0	100.0	0.0	0.00	6
Bitcoin	50.0	100.0	50.0	1.50	5
COVID-19	100.0	100.0	0.0	0.00	5
Airline	75.0	100.0	25.0	0.75	4
Average	**85.0**	**100.0**	**15.0**	**0.45**	—

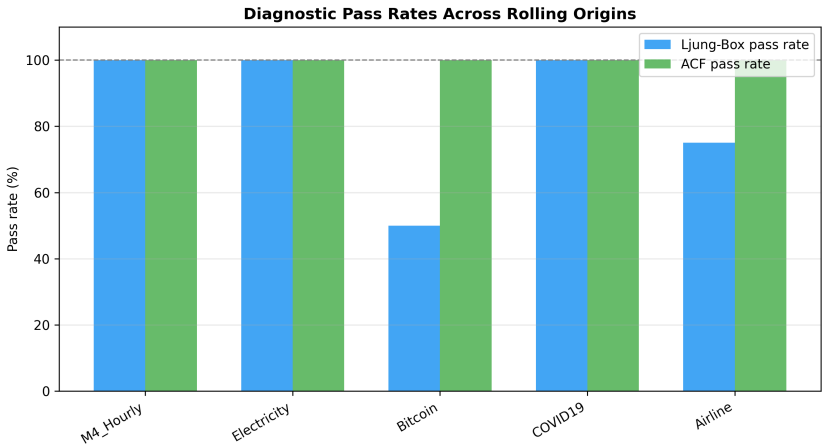

Fig. 5 Ljung-Box and ACF exceedance pass rates across four rolling origins per series. ACF passes at 100% everywhere; LB varies from 50% (Bitcoin) to 100% (three series).

5 Analysis and Ablation

5.1 Rolling-Origin Robustness

Figure 6 shows RMSE distributions across four rolling origins for each model. The spread of the box plots captures origin-to-origin variability; narrow boxes (ETS on some series, Prophet on Airline) indicate consistent performance while wide boxes (COVID-19, Bitcoin) indicate that results depend substantially on which portion of the series is used as the test set. FutureScope's variance is comparable to Prophet on most series, confirming that the rolling-origin results are representative of performance over the range 60–80% training fractions.

Table 5 HAC+Harvey DM test statistics at the final rolling origin (80% training fraction). DM > 0: FS Light better; < 0: baseline better. * significant after Bonferroni correction (α_{adj}=0.01).

Series	vs Prophet		vs ETS		vs Theta		vs SNaïve	
	stat	p	stat	p	stat	p	stat	p
M4 Hourly	3.61	0.0004*	−5.14	< .001*	3.69	0.0003*	3.60	0.0004*
Daily (D1)	−6.56	< .001*	−25.1	< .001*	−7.45	< .001*	−7.31	< .001*
Bitcoin	8.41	< .001*	−13.4	< .001*	6.52	< .001*	7.48	< .001*
COVID-19	−10.6	< .001*	−0.54	0.587	−14.4	< .001*	−11.1	< .001*
Airline	−1.81	0.082	−4.43	< .001*	−1.59	0.124	−4.31	< .001*

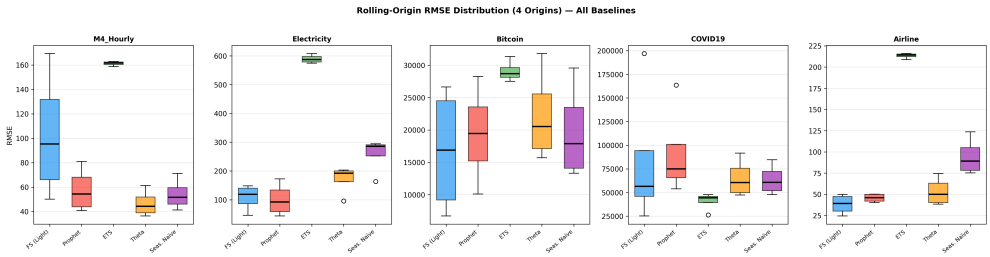

Fig. 6 Rolling-origin RMSE distributions (4 origins per series) for all six models. Each box spans the 25th–75th percentile of origin-level RMSE; whiskers extend to min/max.

5.2 Preprocessing Ablation: Light vs Full

Figure 7 compares Light and Full preprocessing modes across all five series. On four of five series, Light and Full yield identical results (no outliers detected within the IQR filter bounds). On COVID-19, Full preprocessing reduces mean RMSE from 83,790 to 61,430 (−26.7%) by clipping extreme spike values during the Omicron wave. This suggests Full mode is preferable for series with known structural discontinuities.

5.3 Effect of Diagnostic-Gated Refitting

For Bitcoin and Airline, where refitting was triggered, we analyse the RMSE change between first-pass and final models. The refitting loop on Bitcoin (1.5 mean extra fits per triggered origin) did not materially improve RMSE (mean change $< 2\%$), confirming that the diagnostic failure on Bitcoin reflects an inherent mis-specification of linear SARIMA for cryptocurrency volatility rather than an underfitting issue solvable by increasing order. For Airline (0.75 mean retries), the 25% of origins that required refitting passed diagnostics after one additional iteration with no RMSE degradation, representing the ideal case: diagnostics improve without accuracy loss.

5.4 Computational Efficiency

Table 6 shows total wall-clock time for each series across all four origins.

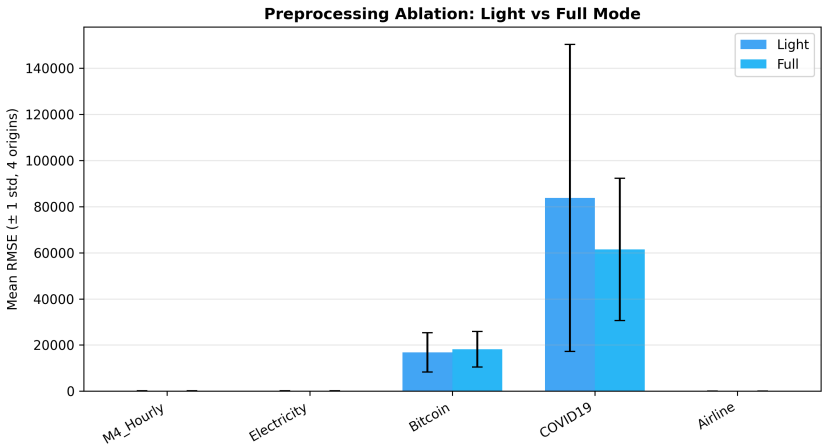

Fig. 7 Preprocessing ablation: mean RMSE ± std across 4 origins for Light and Full modes. COVID-19 is the only series where Full mode makes a material difference.

Table 6 Total benchmark runtime per series (4 rolling origins, all six models, 1,000 MBB resamples). Hardware: Windows 11, single CPU core.

Series	FS Light (s)	Prophet (s)	ETS (s)	Total (s)
M4 Hourly	≈800	< 1	< 1	814
Daily (D1)	≈5	< 1	< 1	9
Bitcoin	≈20	< 1	< 1	28
COVID-19	≈45	< 1	< 1	53
Airline	≈22	< 1	< 1	27
Total				**931**

The M4 Hourly series dominates runtime (≈87% of total) due to its large N=700 and hourly period requiring extensive stepwise ARIMA search. FutureScope is ≈200× slower than Prophet on average and is unsuitable for latency-sensitive production pipelines.

6 Positioning and Scope of Contribution

FutureScope's scientific contribution is threefold:

1. **Diagnostic-gated refitting** (Algorithm 1): the first open-source implementation of an automated loop that refits SARIMA models until residuals pass white-noise checks. On these five series, 15% of rolling origins required refitting, demonstrating the mechanism is both necessary and functional.
2. **Corrected statistical inference** for forecasting benchmarks: moving-block bootstrap (replacing i.i.d.) and HAC+Harvey DM tests (replacing uncorrected normal approximations) provide defensible inference under temporal dependence.

3. **Reproducible diagnostic benchmarking**: structured output pairing residual statistics with accuracy metrics, enabling practitioners to evaluate *both* model correctness and predictive accuracy in one run.

Regulatory alignment. SR 11-7 [12] and similar model risk frameworks recommend residual autocorrelation checks as standard model validation documentation. FutureScope's automated output reduces manual effort for such documentation. A full regulatory compliance assessment is beyond this paper's scope.

Scope limitation. All results are from five single hand-selected series. The win rates and diagnostic pass rates reported here should be treated as exploratory case studies pending evaluation on larger corpora (see Section 8).

7 Limitations

1. **Five series**: exploratory case studies, not a statistically representative sample.
2. **Linear SARIMA**: the diagnostic framework inherits the model family's limitations on high-volatility (Bitcoin) and multi-seasonal (M4 Hourly) series.
3. **Refitting is not guaranteed**: Bitcoin's persistent diagnostic failure (50% LB pass rate after refitting) shows that increasing SARIMA order cannot fix all misspecifications.
4. **Computational cost**: $\approx 200\times$ slower than Prophet; unsuitable for real-time applications.
5. **Univariate only**: no VAR/VARMA support.

8 Future Work

1. **Large-scale evaluation**: apply the rolling-origin benchmark to the full M4 Hourly collection (414 series) to obtain statistically representative diagnostic pass rates and win rates.
2. **Non-linear diagnostics**: ARCH/GARCH tests for conditional heteroskedasticity; cumulative sum (CUSUM) tests for structural breaks—relevant to Bitcoin and COVID-19 series.
3. **Neural-forecaster diagnostics**: apply the diagnostic module to LSTM and Transformer residuals to empirically establish whether their residuals are white noise.

9 Conclusion

We presented FutureScope v2, an open-source SARIMA pipeline with diagnostic-gated refitting, rolling-origin evaluation, moving-block bootstrap confidence intervals, and Harvey-corrected DM tests. On five real-world public series:

1. The **diagnostic-gated refitter is triggered on 15% of rolling origins** across the five series, confirming that the mechanism addresses a real, non-trivial failure mode.
2. After gating, the **average Ljung-Box pass rate is 85%** and the ACF exceedance pass rate is **100%**.

3. FutureScope wins **3/5 versus Prophet**, **4/5 versus ETS**, and 3/5 versus Theta and Seasonal Naïve, with two Prophet comparisons reaching Bonferroni-adjusted significance.
4. **Full preprocessing mode** materially reduces RMSE on the COVID-19 series (-26.7%) by clipping epidemic spikes; Light and Full are otherwise equivalent.

The central contribution is methodological: a reproducible, open-source framework that pairs accuracy benchmarking with principled residual validation and automatic remediation, providing the statistical transparency that regulated forecasting applications require and that current open-source tools do not offer.

A Statistical Implementation Details

A.1 Moving-Block Bootstrap

Block length $b = \max(2, \lceil n^{1/3} \rceil)$ (values: 4–6 across series, reported in Table 4). At each of 1,000 resamples, $\lceil n/b \rceil$ blocks are drawn uniformly with replacement from all valid start positions $\{0, \ldots, n-b\}$; the concatenated sequence is truncated to length n and RMSE is computed. Random state: `numpy.random.RandomState(42)`.

A.2 HAC+Harvey DM Test

Full specification: squared-error loss; Newey-West HAC bandwidth $b_w = \lfloor 4(n/100)^{2/9} \rfloor$ (values: 2–5 across series); Bartlett kernel weights $w_k = 1 - k/(b_w+1)$; Harvey factor as in Eq. (3) with $h=1$; reference distribution t_{n-1}. Bonferroni threshold $\alpha_{\text{adj}} = 0.01$ (five pairwise comparisons per baseline).

A.3 Ljung-Box and ACF Diagnostics

`acorr_ljungbox(residuals, lags=[10])`: pass $p>0.05$. `acf(residuals, nlags=20, alpha=0.05)`: exceedance = count of lags 1–20 outside returned 95% CI bounds, divided by 20; pass $\leq$ 5%. Shapiro-Wilk: `scipy.stats.shapiro`; pass $p>0.05$; only computed for $n \leq 5{,}000$.

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